Design Challenge (compared with wire)

① Wireless channels are a difficult and capacity-limited broadcast communication medium
复杂、容量有限的广播通信媒介

② traffic patterns, user locations and network conditions are constantly changing 变.
业务模式、用户位置和网络条件不断变化

③ Applications are heterogeneous with hard constraints that must be met by the network
应用程序有异构性，网络必须严格满足约束

④ energy and delay constrains change design principles across all layers of the protocol stack
能量、延迟和速率约束会改变协议栈所有层的设计原则
能量和时延限制

History

first wireless：烽火、烟、信鸽

first radio：Marconi

first packet radio：digital，ALOHA

区分
- Cellular：由基站分配到信道　FDMA，CDMA，TDMA　呼叫建立、切换、路由、自适应方案
  小区制的无线通信系统，缩小小区可增加容量，但会加重网络负担
- WLan：随机接入、失败重发

Spectrum Regulation.

FM：87-108MHz.

影响无线科技的进化

Frequency band.

Ultra-high frequency 特高频 300MHz - 3GHz，无线微波链路、蜂窝、地面移动通信网

Super high frequency 超高频 3GHz —　卫星通信

Doppler frequency

物理意义：相对速度 θ∈[$-\frac{\pi}{2}$，$\frac{\pi}{2}$] 为正.

$\Delta d = v\Delta t\cos\theta$　$\Delta\phi = 2\pi v\Delta t\cos\theta/\lambda$　$f_D = \frac{1}{2\pi}\frac{\Delta\phi}{\Delta t} = v\cos\theta/\lambda$

在 free-space 和 ray-tracing models 中忽略，100Hz 太小（1GHz）忽略.

Path Loss

大尺度：较大距离上引起功率的变化

由发射功率的辐射扩散及信道的传播特性造成的。相同的收发距离，path loss 相同.

Free space model.　$\frac{P_r}{P_t} = \left[\frac{\sqrt{G_L}\lambda}{4\pi d}\right]^2$　$P_L\,dB = -10\log_{10}\frac{G_L\lambda^2}{(4\pi d)^2}$　$P_G = -P_L$

例. $f_c = 900MHz$　$d = 10m$．$P_r = 10\mu W$．求 P_t　f_c 变成 5GHz.

$$\frac{P_r}{P_t} = \left(\frac{\sqrt{G_L}\lambda}{4\pi d}\right)^2 \quad P_t = P_r\left(\frac{4\pi d}{\sqrt{G_L}\lambda}\right)^2 = 10^{-5}\left(\frac{4\pi\times 10}{1\times\frac{1}{3}}\right)^2 = 1.42W \cdot \quad P_t = P_r\left(\frac{4\pi d}{\sqrt{G_L}\lambda}\right)^2 = 10^{-5}\left(\frac{4\pi\times 10}{1\times\frac{3}{50}}\right)^2$$

Ray tracing：相比于 LOS 有功率衰减、时延、相移、频移.

two path model：有少量反射体的孤立区域、不适用于室内

delay spread：$\frac{x + x' - l}{c}$

critical distance：$d_c = \frac{4h_t h_r}{\lambda}$

d 较小时，功率随 d^2 个而↓，d 较大时，功率随 d^4 个而↓，与λ独立

Ten-ray model：适用于街道或走廊　d^2

General Ray Tracing：反射、散射、衍射．任意建筑物布局，任意天线位置

Empirical Models

Okumura Model　　1-100km，150-1500MHz，30-100m（基站高度）

HATA　　150-1500MHz，有闭式解

Shadowing

由发射机和接收机之间的障碍物造成的，这些障碍物通过吸收、反射、散射、绕射等方式衰减信号功率，严重时甚至阻断信号

因障碍物数量和类型随机，衰减具有随机性

Multipath

Random # of multipath components, each with varying amplitude, phase, doppler and delay

delay and doppler

path loss and shadowing

Narrowband Channel

nonresolvable multipath components：$|\tau_1-\tau_2| - \frac{1}{B_u}\downarrow\uparrow$　$u(t-\tau_1)\approx u(t-\tau_2)$

接收信号的表达式中，同相分量 $r_I(t)$ 和正交分量 $r_Q(t)$ 是联合高斯随机过程，$\alpha_n(t)$ 瑞利分布，$\phi_n(t)$ 均匀分布

分析时假设没有直射路径（LOS）

lever cross rate：电平通过率 L_Z　单位时间内信号包络向下穿过电平 Z 的平均次数

莱斯分布　K=0 瑞利分布

Average signal fade duration：平均信号衰落时长 $\bar{t}_Z$　信号包络值低于给定电平 Z 的平均时间

$$\bar{t}_Z = \frac{e^{\rho^2}-1}{\rho f_D\sqrt{2\pi}}$$

$\rho=\sqrt{\frac{P_0}{P_r}}$　P_0：接收功率　$\bar{P}_r$：平均接收功率

$T_b \approx \bar{t}_Z$　可能错一位　$T_b \ll \bar{t}_Z$　错多个．　$T_b \gg \bar{t}_Z$　不会错

例1．$P_0 \geq \frac{1}{2}\bar{P}_r$．　$t_Z=60ms$　求 f_D

$$t_Z = \frac{e^{\rho^2}-1}{\rho f_D\sqrt{2\pi}} \qquad \rho=\sqrt{\frac{1}{2}}$$

Wideband Channel.

resolvable multipath components：$|\tau_1-\tau_2| \gg \frac{1}{B_u}\downarrow$．会有ISI．只有抵消 destructive

相比于窄带，加入考虑 multipath delay spread and the time-variations 时延扩展和信道时变性

$c(\tau,t)$　$S_c(\tau,\rho)$ 等效低通冲激响应 $c(\tau,t)$ 的确定散射函数

* characteristics　$A_c(\tau,\Delta t)$　$S_c(\tau,\rho)$

① power delay profile．功率时延谱　$A_c(\tau) \triangleq A_c(\tau,\Delta t=0)$

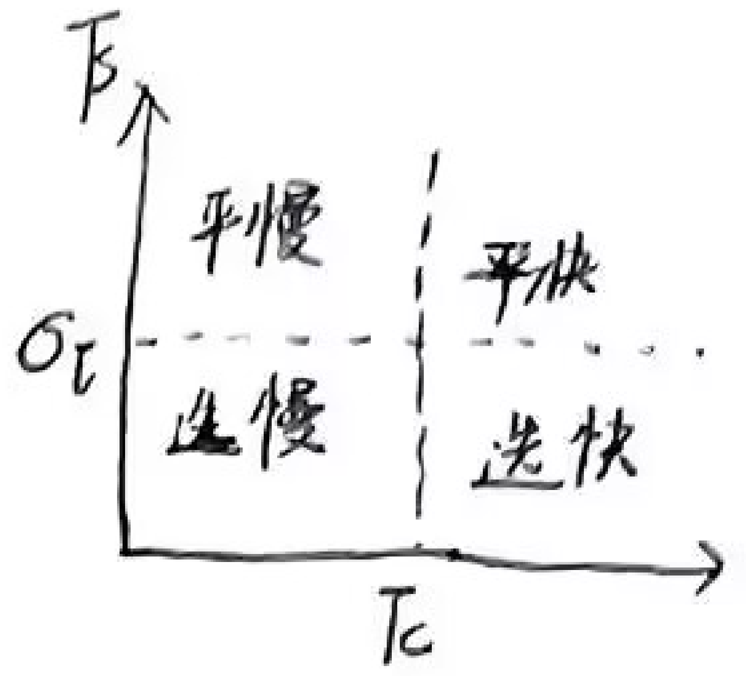
平均时延扩展 μ_{T_m}　average delay spread　$\mu_{T_m} = \frac{\int_0^\infty \tau A_c(\tau)d\tau}{\int_0^\infty A_c(\tau)d\tau}$

均方根时延扩展 σ_{T_m}　rms delay spread　$\sigma_{T_m} = \sqrt{\frac{\int_0^\infty (\tau-\mu_{T_m})^2 A_c(\tau)d\tau}{\int_0^\infty A_c(\tau)d\tau}}$

随机时延扩展 T_m 的pdf　$P_{T_m}(\tau)=\frac{A_c(\tau)}{\int_0^\infty A_c(\tau)d\tau}$

$T_s \ll \sigma_{T_m}$　有严重的ISI．$T_s \gg \sigma_{T_m}$　ISI可忽略．　T_s：线性调制的码元周期

例: $A_c(\tau)=\frac{1}{\bar T_m} e^{-\tau/\bar T_m}$, $\tau>0$ 证 $\mu_{T_m}=\bar T_m$. 并求均方时延扩展 σ_{T_m}



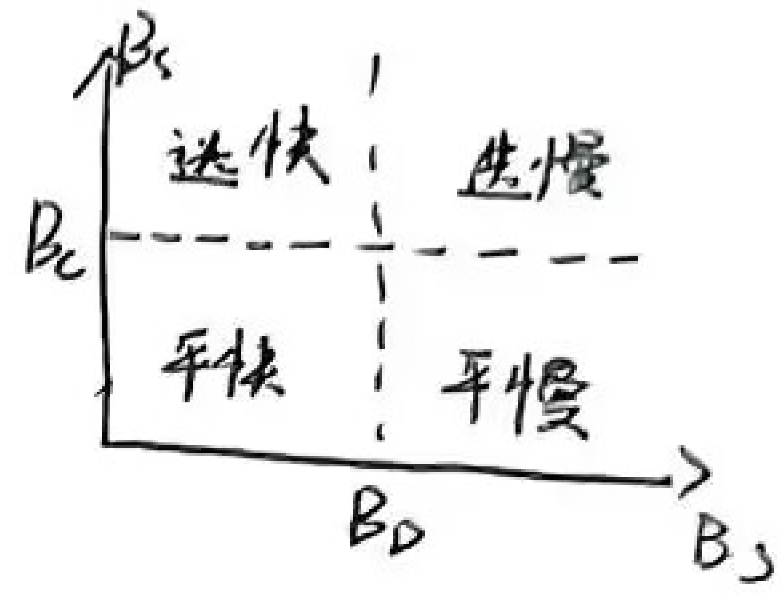
$$\mu_{T_m}=\frac{1}{\bar T_m}\int_0^\infty \tau e^{-\tau/\bar T_m}d\tau=\bar T_m$$

$$\sigma_{T_m}=\sqrt{\frac{1}{\bar T_m}\int_0^\infty \tau^2 e^{-\tau/\bar T_m}d\tau-\mu_{T_m}^2}=2\bar T_m-\bar T_m=\bar T_m$$

② Coherence bandwidth $B_c=\frac{1}{\bar T_m}$

flat fading $B\ll B_c$ $\quad T_s\approx\frac{1}{B}\gg\frac{1}{B_c}\approx\sigma_{T_m}$ 无ISI.

frequency selective fading $B\gg B_c$ $\quad T_s\approx\frac{1}{B}\ll\frac{1}{B_c}\approx\sigma_{T_m}$



③ Doppler power spectrum 多普勒功率谱 $S_c(\rho)$

④ Channel coherence time 信道相干时间 T_c $\quad T_s>T_c$ 快衰落.

使$|S_c(\rho)|$不为零的最大ρ值: 多普勒扩展 Doppler spread. $B_D\approx\frac{1}{T_c}$

fading

各径上相位的快速变化将造成剧烈的干涉现象，从而使接收信号强度发生快速的变化

- flat fading $B\ll B_c$
- frequency selective fading $B\gg B_c$

- fast fading $T_s>T_c$
- slow fading $T_s<T_c$

why digital

开销小、速度快、能效高、易纠错、高频谱效率、抗信道失真、高效的多址接入、更好的安全保密性

basic functions: $\phi_1(t)=\sqrt{\frac{2}{T}}\cos(2\pi f_c t)$ $\quad \phi_2(t)=\sqrt{\frac{2}{T}}\sin(2\pi f_c t)$ $\quad s_i(t)=\sum s_{ij}\phi_j(t)$

例 BPSK $s_1(t)=\alpha\cos(2\pi f_c t)$ $\quad s_2(t)=-\alpha\cos(2\pi f_c t)$ $\quad 0\le t<T$. 求s_{ij}.

$\alpha\cos(2\pi f_c t)=s\sqrt{\frac{2}{T}}\cos(2\pi f_c t)$ $\Rightarrow s_1=\alpha\sqrt{\frac{T}{2}}$ 同理 $s_2=-\alpha\sqrt{\frac{T}{2}}$

signal constellation 信号星座图 decision region 判决区间

例 BPSK 对应星座点 $s_1=A$、$s_2=-A$ $(A>0)$. 求判决域 Z_1、Z_2

$Z_1=\{r:\|r-A\|<\|r-(-A)\|\}=\{r:r>0\}$ $\quad Z_2=\{r:\|r-(-A)\|<\|r-A\|\}=\{r:r<0\}$.

取决于r_s
对于一个点，能取到d_{min}的个数

P_e the union bound $P_e\le\frac{1}{M}\sum_{i=1}^{M}\sum_{k=1,k\ne i}^{M}Q(\frac{d_{ik}}{\sqrt{2N_0}})$ $\quad$ looser bound $P_e\le(M-1)Q(\frac{d_{min}}{\sqrt{2N_0}})$

closed form $P_e\le\frac{M-1}{\sqrt{\pi}}\exp[\frac{-d_{min}^2}{4N_0}]$ $\quad$ nearest neighbor approximation $P_e\approx M_{d_{min}}Q(\frac{d_{min}}{\sqrt{2N_0}})$

例: 四个星座点 $s_1(A,0)$ $s_2=(0,A)$ $s_3=(-A,0)$ $s_4=(0,-A)$ 设 $A/\sqrt{N_0}=4$、求星座图的最小距离和P_e.

$d_{min}=d_{12}=d_{23}=d_{34}=d_{41}=\sqrt{2A^2}$. 其他的 $d=2A$

对称性，只考虑一个

the union bound $P_e\le\sum_{j=2}^{4}Q(\frac{d_{1j}}{\sqrt{2N_0}})=2Q(\frac{A}{\sqrt{N_0}})+Q(\frac{\sqrt{2}A}{\sqrt{N_0}})=2Q(4)+Q(\sqrt{32})$

looser bound $P_e\le(M-1)Q(\frac{d_{min}}{\sqrt{2N_0}})=3Q(4)$

closed form $P_e\le\frac{3}{\sqrt{\pi}}\exp[\frac{-0.5A^2}{N_0}]$

nearest neighbor approximation $P_e=M_{d_{min}}Q(\frac{d_{min}}{\sqrt{2N_0}})=2Q(4)$

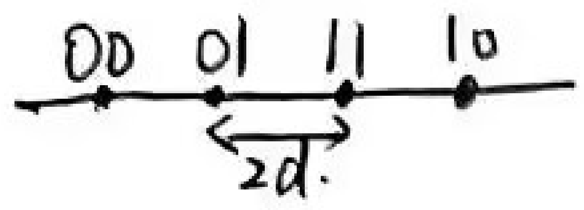
Linear Modulation (MPAM. MPSK. MQAM)

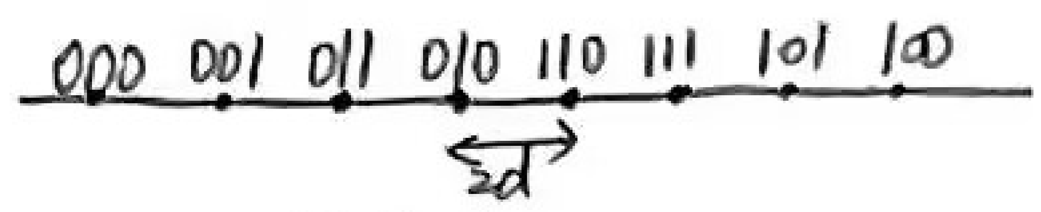
MPAM $d_{min}=2d$. 平均能量 $E_s=\frac{1}{M}\sum_{i=1}^{M}A_i^2$ $\quad A_i=(2i-1-M)d$.

例: 4PAM $A_i=(2i-1-M)d=(2i-5)d$. $\quad A_i\in\{-3d,-d,d,3d\}$

$$E_s=\frac{1}{4}(9d^2+d^2+d^2+9d^2)=5d^2$$

Gray encoding：MPSK和方形MQAM可，矩形星座的MQAM不可。





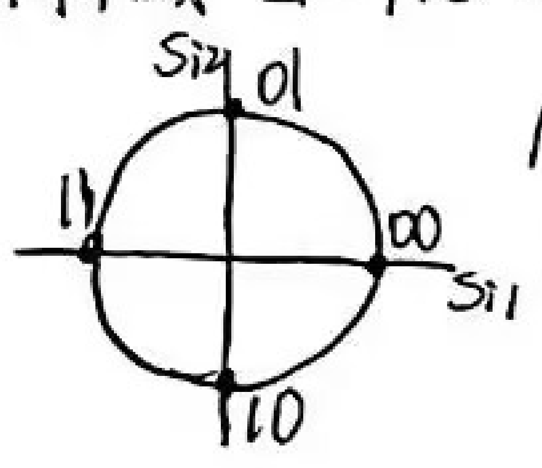
$M=4,\ k=2$　　　　$M=8,\ k=3$

判决域 $Z_i=\begin{cases}(-\infty, A_i+d) & i=1\\ [A_i-d, A_i+d] & 2\le i\le M-1\\ [A_i-d, +\infty) & i=M\end{cases}$

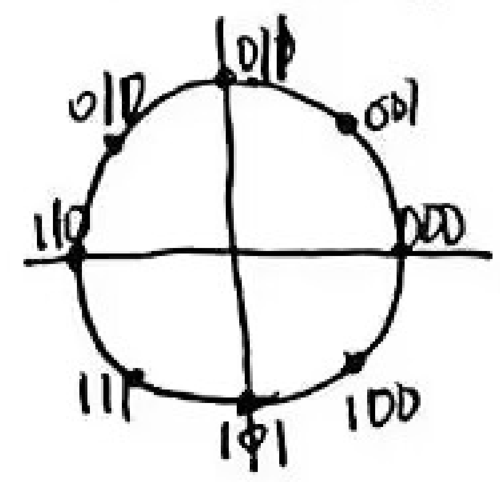
MPSK　$d_{min}=2A\sin(\frac{\pi}{M})$

每个符号有相同的能量 $E_{si}=\int_0^{T} s_i^2(t)\,dt=A^2$.

判决域 $Z_i=\{re^{j\theta}: 2\pi(i-1.5)/M\le\theta<2\pi(i-0.5)/M\}$

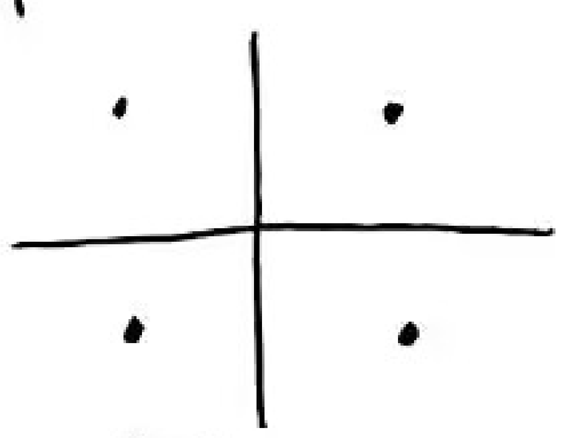


$M=4$
$K=2$

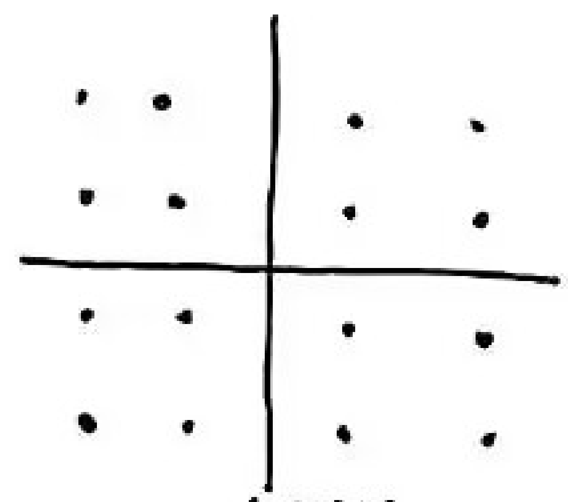


$M=8$
$K=3$

MQAM



4QAM



16QAM.

~~DQ~~DPSK.

例 DPSK 101110 $s(k-1)=Ae^{j\pi}$

1：改变180°　$s(k)=A$　　　　下面 A、$Ae^{j\pi}$、$Ae^{j\pi}$

0：不变　$s(k+1)=A$　　　　综上 A. A. $Ae^{j\pi}$. A. $Ae^{j\pi}$. $Ae^{j\pi}$

1：变180°　$s(k+2)=Ae^{j\pi}$

同理，也要会 DQPSK，每次变90°

优：对载波相位的随机漂移并不敏感

缺：①若有显著的多普勒频移，相邻码元间隔内的相位可能会不相关，使得前一个符号成为一个很差的相位参考

② irreducible error floor

Nonlinear modulation (FSK)

最低频率间隔　FSK：$2\Delta f_c$　　相干解调难度大、成本高。

MSK：$2\Delta f_c=\frac{1}{2T_s}$　满足正交性的最小间隔，MSK是最小带宽的FSK

(对比) 线性调制有更好的频谱特性，易受衰落和干扰的影响，频率调制可用价格便宜、功率更高的非线性放大器

SNR

$$SNR=\frac{P_r}{N_0B}=\frac{E_s}{N_0BT_s}=\frac{E_b}{N_0BT_b}$$

若成形脉冲 $T_s=\frac{1}{B}$　~~X~~ 符号信噪比 $\gamma_s=\frac{E_s}{N_0}$　比特信噪比 $\gamma_b=\frac{E_b}{N_0}$

$$\gamma_b=\frac{\gamma_s}{\log_2 M}\qquad P_b=\frac{P_s}{\log_2 M}$$

BPSK: $P_b = Q(\sqrt{2\gamma_b})$

QPSK: $P_s \approx 2Q(\sqrt{\gamma_s})$ the nearest neighbor approximation

例: γ_b=7dB,求格雷码的QPSK的 P_b. P_s。比较P_b的精确值与$P_b=\frac{P_s}{2}$得到的近似值

$P_b = Q(\sqrt{2\gamma_b})$ 精确: $P_s = 1-[1-Q(\sqrt{2\gamma_b})]^2$ 近似 $P_s = 2Q(\sqrt{\frac{\gamma_s}{2}})$

MPSK: $P_s = 2Q(\sqrt{2\gamma_s}\sin(\frac{\pi}{M}))$

例: γ_b=15dB 时 8PSK和16PSK的 P_b.

通过γ_b求 $\gamma_s = \gamma_b \log_2 M$, 有$\gamma_s$可求 $P_s = 2Q(\sqrt{2\gamma_s}\sin(\frac{\pi}{M}))$. 根据 $P_b = \frac{P_s}{\log_2 M}$ 求P_b

MPAM: $P_s = \frac{2(M-1)}{M} Q(\sqrt{\frac{6\gamma_s}{M^2-1}})$

MQAM: $P_s \approx M_{d_{min}} Q(\frac{d_{min}}{\sqrt{2N_0}})$

Fading

标准: ①中断率 P_{out} (The outage probability). γ_s低于某值的概率.对应于可容忍的最大误码率 $T_s \ll T_c$

②平均错误率 $\bar{P}_s$ (The average error probability). 针对γ_s取平均 $T_s \approx T_c$

③.二者结合.一定时间或空间内的平均错误率

当 slow fading 低于某值时,发生中断.非中断时性能为对fast fading取平均.

$\bar{\bar{\gamma}}_s$: path loss 为定值,对 fast fading 和 shadowing 取平均的符号信噪比

$\bar{\gamma}_s$: path loss 和 shadowing 为定值.对fast fading —

γ_s: path loss. shadowing. fast fading 均为定值

Diversity

定义:在独立的衰落路径上发送相同的数据,由于独立路径在同一时刻经历深衰落的概率很小.因此经过适当的合并后,接收信号的衰落程度就会被减小

实现 ①space diversity 空间分集

②.Antennas with different polarization 使用不同极化方式的天线

③.Directional antennas . smart antenna 定向天线.智能天线

④.time diversity 时间分集

combining techniques.

①.selection combining 选择合并:输出信噪比最高的

②.Maximal Ratio combining 最大比合并 加权和

③.Equal gain combining 等增益合并:以相同的权重进行同相合并.

④门限合并.

Channel coding with interleaving

交织分散错误、编码负责纠错

Duplex

In a cellular system, duplexing is the key technique in air interface. to guarantee the bi-directional transmission between downlink and uplink.

FDD (frequency domain duplexing)

cheap duplex filters are sufficient for achieving very good seperation between the uplink and downlink.

TDD

①. We can flexibley allocate the number of uplink and downlink slots, which is very useful for unbalanced application.

②. the uplink and downlink channels are reciprocal: the uplink and downlink use the same frequency band and the duplexing time is much smaller than coherence time

dis: limited by time delay or cell coverage.

Multiple Access

FDMA:

flat fading channel and no serious ISI occurs: the ~~bandwith~~ bandwidth of each FDMA subchannel is relatively narrow (30kHz), which is less than the channel coherence bandwidth.

ICI: the carrier frenquency synchronization will be sensitive to the frequency offset.

载波频率同步受频偏的影响.

TDMA.

①. TDMA/TDD mode: reciprocity can be used to improve the efficiency of channel estimation.

②. 根据不同用户的服务类型和服务质量的要求、为他们分配不同数量的时隙

dis: works at bursts mode: 每个时隙都需要同步.

CDMA.

①. 接收端通过与PN码做相关运算提取期望码字. 由于正交性, 与其他非期望码字的输出可视为噪声.

②. soft capacity limit: 存在足够数量的正交PN码, 可按需增加用户数量

dis: the near-far problem: 远离基站的信号比靠近基站的信号要弱, 强信号会提升弱信号的噪声水平. 使弱信号解调更难. 解决: channel inversion 翻转

ALOHA: pure ALOHA. Slotted-ALOHA、Reservation-ALOHA.

Channel Planning P304

the cluster size (簇): $N = i^2 + ij + j^2$ ① 通常大于1. CDMA可=1. 通过码分的频率和时隙可以相同

~~normalizat~~ nomalized reuse distance $D_{norm} = \frac{D}{R} = \sqrt{3N}$ $D = \sqrt{3N} R$.

例: ~~每~~ TDMA、每个信道带宽 200kHz. SIR 10dB fading margin 10dB

$\frac{D-R}{R} = (100)^{\frac{1}{4}} = 3.16$ $\frac{D}{R} = 4.16$ $D = \sqrt{3N} R. \Rightarrow N \geq 5.76$ … 取整 $N = 7$ (20dB=100倍)

问题: handoff 越区切换 CDMA soft handoff TDMA. FDMA hard handoff